

EBU Recursive Möbius Growth Theorem and Study Bridge

Purpose and status

This document records a prospective mathematical and scientific bridge between:

- growing population, resources, and demand;
- expansion of a declared EBU dependency topology;
- canonical recursive motifs;
- Boolean or feasible-poset Möbius decomposition;
- certified reuse of previously solved structures;
- correction and cache-invalidation propagation; and
- long-run viability and homeostasis tests.

It is a conceptual input for a separately audited Stage D authority clarification. It does not modify the accepted EBU equation, authorize a Stage E harness, execute a Stage F study, report a scientific result, or revise a book manuscript.

The central claim is not that Fibonacci is a universal law. The central claim is that a declared recursive composition admits an exact Möbius interaction identity, and that certified self-similarity may permit exact reuse of earlier calculations.

1. Three objects that must remain distinct

The programme joins three mathematical objects without conflating them.

1. **Population and demand dynamics** determine how the requested scale changes through time.
2. **Topology and motif growth** determine how providers, actions, resources, constraints, and their declared dependencies are expanded.
3. **Möbius interaction and inverse reconstruction** determine how individual, pairwise, and higher-order contributions reconstruct the declared EBU set function.

Population growth does not select a topology automatically. A provider, institution, scientific domain, or model authority declares or measures the topology. EBU validates its typed, versioned, provenance-linked, and boundary-complete use.

2. Time- and topology-indexed EBU without changing the core equation

Let

$$\Theta_t$$

be the declared topology at time t , and let A_t be its finite action set. Write the existing EBU set function with its context exposed:

$$E_t(S) = E(S; \Theta_t, x_t, h_t, \pi_t), \quad S \subseteq A_t,$$

where x_t is the declared state, h_t the required history and boundary information, and π_t the frozen protocol.

This indexing does not add a Fibonacci term to the EBU equation. It makes explicit that a value belongs to a particular time, topology, state, history, and protocol.

3. Raw Möbius decomposition and inverse reconstruction

For a finite Boolean action domain with an explicit empty baseline,

$$I_t(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} E_t(T).$$

The inverse, or zeta reconstruction, is

$$E_t(S) = \sum_{T \subseteq S} I_t(T).$$

The empty coefficient remains

$$I_t(\emptyset) = E_t(\emptyset).$$

For a declared finite feasible poset P_t , the corresponding incidence algebra formulas are

$$I_t(x) = \sum_{y \leq x} \mu_{P_t}(y, x) E_t(y), \quad E_t(x) = \sum_{y \leq x} I_t(y).$$

No Boolean value may be invented for an infeasible subset.

4. Recursive Möbius surplus identity

Assume a declared ordered recursive motif family

$$M_{n+1} = M_n \oplus M_{n-1}.$$

All compared values must use the same units, empty baseline, boundary, initial augmented state, history, feasibility, removal, preservation, resolver, loss, process-account, commitment, and settlement semantics.

Define

$$R_n = E(M_n) - E(\emptyset).$$

At level n , treat the two child motifs as two macro-actions a and b :

$$e_n(\emptyset) = E(\emptyset),$$

$$e_n(\{a\}) = E(M_n), \quad e_n(\{b\}) = E(M_{n-1}),$$

$$e_n(\{a, b\}) = E(M_{n+1}).$$

The raw two-child Möbius coefficient is

$$\begin{aligned} I_n(\{a, b\}) &= e_n(\{a, b\}) - e_n(\{a\}) - e_n(\{b\}) + e_n(\emptyset) \\ &= R_{n+1} - R_n - R_{n-1}. \end{aligned}$$

Therefore the accepted recursive surplus is exactly the macro-level Möbius interaction:

$$\boxed{J_n = I_n(\{a, b\}) = R_{n+1} - R_n - R_{n-1}}.$$

Inverse Möbius reconstruction gives the exact forced recurrence

$$\boxed{R_{n+1} = R_n + R_{n-1} + J_n}.$$

This is an algebraic theorem under the stated common-protocol assumptions. It is not a scientific observation about a real population.

5. Interpretation of the recursive interaction

- $J_n = 0$ means same-protocol structural additivity at level n .
- $J_n > 0$ means a positive recursive interaction under that protocol.
- $J_n < 0$ means a negative recursive interaction under that protocol.

The sign alone does not prove causality, fairness, cooperation, welfare, or universal social benefit. It identifies the non-additive term required for exact reconstruction.

6. Fibonacci projection with interaction forcing

Let $F_0 = 0$ and $F_1 = 1$. Iterating the forced recurrence gives, for $n \geq 1$,

$$\boxed{R_n = F_n R_1 + F_{n-1} R_0 + \sum_{k=1}^{n-1} F_{n-k} J_k}.$$

Proof

The result is true at $n = 1$. At $n = 2$, it is exactly

$$R_2 = R_1 + R_0 + J_1.$$

Assume the formula for R_n and R_{n-1} . Substituting both expressions into

$$R_{n+1} = R_n + R_{n-1} + J_n$$

and using

$$F_{m+1} = F_m + F_{m-1}$$

combines the base terms and every earlier forcing term into the required formula at $n + 1$.

Consequence

When every $J_k = 0$, the surplus sequence obeys a homogeneous Fibonacci-form recurrence. Fibonacci structure is not inferred merely because this special case is mathematically available.

7. Lucas projection

Lucas numbers satisfy

$$L_{n+1} = L_n + L_{n-1}, \quad L_0 = 2, \quad L_1 = 1.$$

If every $J_k = 0$ and

$$R_0 = 2c, \quad R_1 = c,$$

then

$$\boxed{R_n = cL_n}.$$

Lucas behavior is therefore an initial-condition family of the same second-order recursion. It is a useful registered control, not a separate universal EBU law.

8. Exact missing-interaction error and correction propagation

If a model reconstructs the recursive system while incorrectly setting all interactions to zero, its projection is

$$\hat{R}_n = F_n R_1 + F_{n-1} R_0.$$

The exact omitted-interaction error is

$$\boxed{R_n - \hat{R}_n = \sum_{k=1}^{n-1} F_{n-k} J_k}.$$

A correction δ to one earlier interaction J_q produces

$$\boxed{\Delta R_n = F_{n-q} \delta, \quad n > q}.$$

This is dependency propagation through the declared recurrence. It is not a physical wave, phase interference, or evidence of causal transmission.

For a general Boolean table whose raw values change by $\Delta E(Q)$, linearity gives

$$\Delta I(S) = \sum_{Q \subseteq S} (-1)^{|S|-|Q|} \Delta E(Q).$$

For a feasible poset,

$$\Delta I(x) = \sum_{q \leq x} \mu_P(q, x) \Delta E(q).$$

These formulas define the algebraic affected region. Operational cache reuse still requires complete dependency, alias, boundary, history, authority, and provenance validation.

9. Recursive coefficient transport and scaling

An occurrence reused at a larger recursive level is not necessarily coefficient-identical to its smaller source. Canonical identity answers whether two declared structures are equivalent under the accepted encoding. It does not by itself say whether an extensive, intensive, boundary, pairwise, or higher-order coefficient is unchanged when the system is enlarged.

Before comparing coefficients across levels, declare a role-preserving transport map

$$\iota_n : \mathcal{I}(M_n) \longrightarrow \mathcal{I}(M_{n+1})$$

from exposed interaction coordinates of the smaller motif to their corresponding coordinates in the larger motif. Then declare or derive a typed scaling operator

$$\mathcal{S}_{n,S}$$

for every transportable coordinate. The simplest scalar case is

$$\hat{I}_{n+1}(\iota_n S) = \lambda_{n,S} I_n(S).$$

The exact scaling residual is

$$\boxed{K_{n+1}(\iota_n S) = I_{n+1}(\iota_n S) - \lambda_{n,S} I_n(S)}.$$

- $K_{n+1} = 0$ establishes the declared exact scaling relation for that coordinate and protocol.
- $K_{n+1} \neq 0$ records new interaction, boundary, state, congestion, delay, or other scale-dependent content that cannot be obtained by scaling alone.
- An undeclared $\lambda_{n,S}$, missing transport map, changed units, or changed query domain forbids scaled reuse.

Several scaling classes must remain distinct:

1. **Invariant/intensive:** $\lambda_{n,S} = 1$.
2. **Occurrence-extensive:** $\lambda_{n,S} = N(M_{n+1})/N(M_n)$, but only when a theorem or domain declaration proves identical per-occurrence contribution and complete additivity for that coefficient.
3. **Boundary-extensive:** scaling follows the declared exposed-boundary measure, not total occurrence count.
4. **Degree-homogeneous:** $\lambda_{n,S} = s_n^{\alpha_S}$ for a separately justified scaling degree α_S .
5. **Non-scalable:** direct recomputation is required.

For the declared Fibonacci occurrence family,

$$N(M_n) = F_{n+1},$$

so an occurrence-extensive coefficient has the prospective scale ratio

$$\boxed{\lambda_n^{\text{occ}} = \frac{F_{n+2}}{F_{n+1}} \longrightarrow \varphi.}$$

This limit does not imply that every EBU or Möbius coefficient scales by the golden ratio. Pairwise and higher-order interactions may scale with boundary size, interface count, density, congestion, or not obey a closed scaling law at all. Lucas-compatible occurrence families have an analogous consecutive-ratio limit but different finite-level initial conditions.

When a smaller coefficient is corrected by $\Delta I_n(S)$, the transported larger coefficient obeys

$$\Delta I_{n+1}(\iota_n S) = \lambda_{n,S} \Delta I_n(S) + \Delta K_{n+1}(\iota_n S).$$

Therefore a scaling rule becomes part of the cache key, dependency DAG, and invalidation receipt. A changed factor, scaling degree, transport map, boundary, or residual invalidates every dependent larger-level result.

The scientific programme must compare direct expanded coefficients with scaled predictions at every tested level. It must include exact invariant, extensive, boundary-scaled, deliberately misscaled, and non-scalable controls. Scaling factors belong to mathematical derivation or provider/domain/model authority; EBU does not infer the socially or physically correct factor automatically.

10. Incremental topology-extension theorem

Let the action domain expand from

$$A_t \quad \text{to} \quad A_{t+1} = A_t \cup B_t.$$

For an old subset $S \subseteq A_t$, if

$$E_{t+1}(T) = E_t(T) \quad \text{for every } T \subseteq S,$$

then

$$\boxed{I_{t+1}(S) = I_t(S)}.$$

This follows immediately because the defining Möbius sum is identical term by term. New elements do not require old coefficients to be recomputed unless they change old subset values, boundaries, histories, feasibility, or other result-affecting authority.

If an old subset value changes, its correction propagates through the declared upward cone. If a certified boundary is crossed or invalidated, all dependent summaries must be reopened. Stale reuse must fail closed.

11. Population and demand coupling

Population is a model input, not an EBU-discovered topology. One registered discrete logistic scenario may be

$$P_{t+1} = P_t + rP_t \left(1 - \frac{P_t}{K}\right),$$

with demand

$$D_t = q_t P_t + d_t^{\text{institutional}} + d_t^{\text{shock}}.$$

The scientific programme must also include linear, exponential, seasonal, pulsed, shock, declining, migration-like, and adversarial demand controls. Logistic growth is neither assumed nor fitted after outcomes are visible.

Given declared capacity C_n and reserve margin m_t , an expansion policy may choose

$$n_t = \min\{n : C_n \geq D_t + m_t\}.$$

If, under separately proved additive-capacity assumptions,

$$C_n = cF_{n+1},$$

then

$$n_t = \Theta(\log D_t).$$

For fixed-carrying-capacity logistic demand, the required level eventually saturates. This is a conditional capacity projection. It does not prove homeostasis, timely construction, adequate resources, or social desirability.

11.1 Bidirectional capacity-population hypothesis

The demand-driven direction above is only one registered causal arm. The programme must also test the reverse and bidirectional possibilities:

recursive expansion $\longrightarrow$ net interaction benefit $\longrightarrow$ usable capacity $\longrightarrow$ possible demographic response.

For a capacity-compatible transported coefficient, define the net capacity created at expansion $n + 1$ only after all construction, maintenance, resource, correction, delay, and loss accounts have been settled:

$$\boxed{C_{n+1} = \lambda_n C_n + K_{n+1}^C - X_{n+1}},$$

where K_{n+1}^C is the interaction or scaling residual that a declared physical model permits to be converted into usable capacity, and X_{n+1} is the complete expansion burden. An abstract positive EBU coefficient is not automatically capacity and must not be relabelled as such.

The demographic bookkeeping remains separately declared:

$$P_{t+1} = P_t + B_t - D_t^{\text{death}} + M_t,$$

with births, deaths, and migration allowed to depend on the audited capacity, service, health, reserve, environmental, institutional, and behavioural state. Therefore positive net capacity can enable population growth without implying that it universally causes population growth. The benefit may instead appear as

reserves, resilience, lower burden, improved quality, regeneration, or a change in demographic behaviour.

For the special two-child recursive capacity law

$$C_{n+1} = C_n + C_{n-1} + J_n^C - X_{n+1},$$

if a separately declared demographic response keeps population at $P_n = \rho C_n$, then the following conditional projection is exact:

$$\boxed{P_{n+1} = P_n + P_{n-1} + \rho(J_n^C - X_{n+1})}.$$

This is a testable bridge, not a universal demographic law. It states exactly which assumptions would allow a recursively generated capacity surplus to support a Fibonacci-forced population-capacity path.

Stage F must distinguish three preregistered causal arms:

1. population or demand drives topology expansion;
2. topology-generated net capacity permits a demographic response; and
3. population, demand, topology, capacity, and interaction residuals form a bidirectional feedback loop.

Controls must include zero and negative K^C , positive residuals that are not capacity-convertible, benefits allocated to reserves or quality rather than population, demographic transition, delay, shocks, and nonrecursive topologies. Only independently audited evidence may determine which direction, if any, is supported.

12. Conditional computational-compression corollary

For the declared ordered Fibonacci substitution family,

$$N(M_n) = F_{n+1} = \Theta(\varphi^n).$$

Under all accepted A1–A8 boundary-compression conditions, uniformly bounded summary size, and uniformly bounded composition cost, summaries through level n require $O(n)$ new compositions. Therefore

$$\boxed{T_{\text{summary}} = O(\log N)}$$

in the expanded occurrence count N .

This does not make physical construction, measurement, maintenance, storage, or arbitrary all-subset reconstruction logarithmic. Without valid structure, exact arbitrary Boolean Möbius analysis still requires all 2^N subset values, with a fast transform costing approximately $O(N2^N)$ arithmetic operations plus subset-evaluation cost.

13. Connection to homeostasis

Growing-system homeostasis must be tested over a moving model, not inferred from topology compression.

Let

$$\mathcal{V}_t = \mathcal{V}(P_t, D_t, \Theta_t)$$

be the registered viable region. The trajectory must demonstrate, under the exact registered conditions,

$$x_t \in \mathcal{V}_t$$

for the declared horizon while accounting for:

- construction and activation delay;
- resource acquisition and depletion;
- reserve margins;
- unmet demand;
- service continuity;
- regeneration;
- congestion and interaction burdens;
- correction actions and receipts;
- shocks and recovery;
- failed or delayed expansion; and
- computational delay in detecting and applying changes.

Topology compression can make adaptation computationally cheaper. It does not guarantee sufficient physical capacity or homeostasis.

14. Registered study design

Stage D: freeze the questions

The authority should join the existing SD-01, SD-06, SD-07, SD-10, SD-11, and SD-14 programmes without replacing their accepted evidence:

- SD-01: growing population/demand and moving viability;
- SD-06: direct Möbius and inverse/zeta reconstruction oracle;
- SD-07: recursive Möbius surplus, Fibonacci/Lucas controls, canonical reuse, and incremental invalidation;
- SD-10: delayed correction and computational-feedback effects;
- SD-11: complete correction and conservation receipts; and
- SD-14: integrated growing multi-domain scenario.

Stage E: implement immutable oracles and the harness

Before inspecting scientific outcomes, implement:

- direct expanded evaluation;
- direct Boolean and feasible-poset Möbius oracles;

- inverse/zeta reconstruction;
- recursive reconstruction using recorded J_n ;
- canonical-equivalence and A1–A8 certificate validation;
- complete cache keys;
- deterministic invalidation and correction receipts;
- population/demand generators;
- growing-topology generators;
- checkpoint continuation; and
- scientific and computational counters.

Stage F: execute the registered campaign

Required topology families:

- Fibonacci recursive motifs;
- Lucas-compatible initial conditions;
- other second-order recursive motifs;
- non-Fibonacci substitution systems;
- balanced trees and chains;
- random-recursive structures;
- perturbed near-motifs;
- nonrecursive random topologies; and
- adversarial broad-dependency topologies.

Required demand families:

- fixed;
- linear;
- exponential;
- logistic with declared carrying capacity;
- seasonal or pulsed;
- shocks;
- declining population; and
- adversarial demand.

Required computational arms:

1. complete reconstruction;
2. incremental extension without motif reuse;
3. certified canonical-motif reuse;
4. deliberately incomplete cache key;
5. stale-cache correction control;
6. false canonical-equivalence control; and
7. broad reconfiguration where local reuse should provide little or no gain.

Required equivalence condition

For every positive reuse disposition,

$$R_n^{\text{recursive}} = R_n^{\text{fully expanded}}$$

under the exact declared numerical policy. A mismatch is a failure, not an approximation.

Scientific measurements

- viability-set membership;
- homeostasis under the registered definition;
- delivered and unmet demand;
- reserves and regeneration;
- recovery time and collapse;
- cumulative burden and loss;
- interaction and correction receipts;
- conservation residuals; and
- sensitivity to expansion delay and topology error.

Computational measurements

- wall time and CPU time;
- peak memory;
- subset and hyperedge evaluations;
- direct versus reused coefficients;
- canonicalization work;
- cache-hit rate;
- invalidation radius;
- propagation depth;
- storage and trace size; and
- checkpoint/restart accounting.

15. Falsifiers

The programme must reject a positive theorem or scientific disposition if:

- recursive reconstruction differs from complete reconstruction;
- a transported coefficient uses an undeclared or empirically selected scaling factor;
- a coefficient is scaled by occurrence count despite being intensive, boundary-dependent, or non-scalable;
- a nonzero scaling residual is hidden rather than recorded and propagated;
- a reused motif lacks any A1–A8 obligation;
- an old coefficient is reused after a result-affecting change;

- invalidation omits a dependent parent or alias;
- Fibonacci or Lucas behavior is selected only after outcomes are inspected;
- a non-Fibonacci control performs equally while the interpretation claims Fibonacci specificity;
- computational work is hidden or measured on incomparable implementations;
- physical construction cost is mislabeled as summary-computation cost;
- topology compression is reported as homeostasis;
- capacity is silently created without a resource or action receipt; or
- a positive interaction or capacity residual is said to ensure population growth without an independently declared and tested demographic response;
- positive interaction is presented as causality, fairness, or social benefit.

16. Nature as motivation, not authority

Biological growth provides a valuable explanatory analogy because many living structures expand through repeated local construction rather than demolition and complete reconstruction.

Sunflower and related flower heads often exhibit clockwise and anticlockwise parastichy counts that are consecutive Fibonacci numbers. Controlled research also documents non-Fibonacci and irregular heads. This is a useful accessible book reference showing why visual resemblance must not be treated as a universal law. It is not an EBU authority source and does not determine the mathematical theorem, topology programme, controls, or scientific interpretation.

Useful sources:

- Swinton et al., “Novel Fibonacci and non-Fibonacci structure in the sunflower,” *Royal Society Open Science* (2016): <https://pmc.ncbi.nlm.nih.gov/articles/PMC4892450/>
- Zhang et al., “Phyllotactic patterning of gerbera flower heads,” *Proceedings of the National Academy of Sciences* (2021): <https://pmc.ncbi.nlm.nih.gov/articles/PMC8020676/>

Horns and other pointed biological structures can display curved, spiral, or power-law growth generated by differential growth. The stronger statement that ram horns generally grow according to the Fibonacci sequence is not established well enough to serve as programme evidence.

- Evans et al., “A universal power law for modelling the growth and form of teeth, claws, horns, thorns, beaks, and shells,” *BMC Biology* (2021): <https://pmc.ncbi.nlm.nih.gov/articles/PMC8008625/>

The defensible analogy is therefore:

Local generative rules can add new structure while preserving and reusing earlier organization. EBU tests when an analogous declared recursive organization preserves exact results, reduces recomputation, and remains physically viable under growing demand.

17. Book integration without chapter bloat

No book manuscript should be revised before accepted Stage F execution and Stage G result audit. The future structure should reserve these placements:

- **Part V:** population, demand growth, moving viability, reserves, expansion delay, homeostasis, and the bidirectional capacity-population hypothesis;
- **Part VI:** recursive Möbius surplus theorem, Fibonacci/Lucas projections, coefficient transport and scaling residuals, canonical motifs, inverse reconstruction, and certified reuse;
- **Part VIII:** a short cross-reference explaining correction propagation, invalidation, and feedback delay;
- **Mathematical appendix:** full proofs of the recursive surplus identity, forced Fibonacci projection, Lucas corollary, coefficient-transport and conditional-scaling results, incremental extension theorem, poset generalization, and complexity bounds; and
- **Results chapters:** only independently audited Stage F/G figures, tables, run identities, and code/configuration references.

The nature comparison may appear briefly as an accessible supporting example. It must not be presented as the scientific basis of the programme and must not replace the theorem, registered controls, or EBU evidence.

The relevant book sequence should culminate in one light synthesis section, not a disconnected additional chapter:

Recursive topology may generate net interaction benefits that expand usable carrying capacity. Where demographic behaviour responds positively to that capacity, the benefit can enable population growth. EBU tests this connection rather than presuming it universally.

This conclusion may be expanded into a scientific finding only after Stage F execution and Stage G audit. Before then it must be labelled as the central conditional hypothesis. The accompanying reader explanation may state that a successful system could help create the physical and institutional conditions under which future generations have meaningful resources available; it must not promise, guarantee, or centrally determine demographic outcomes.

18. Feature value for the reader

This programme gives EBU a precise layered architecture:

- the declared topology states where interactions may exist;
- Möbius coefficients quantify non-additive interactions;
- inverse reconstruction rebuilds the complete declared value;
- recursive motifs describe how larger structures are assembled;
- typed scaling operators describe how eligible coefficients change with scale;
- scaling residuals expose what cannot be inherited from the smaller motif;
- canonical certificates determine when earlier calculations may be reused;
- invalidation propagates corrections to every dependent larger structure;
- population and demand models drive when expansion is needed; and
- net recursive capacity can also be tested as a possible enabler of later demographic growth, completing the bidirectional feedback hypothesis; and

- long-run tests determine whether the growing system remains viable.

The milestone is not a universal Fibonacci system. It is an exact way to test whether recursive structure turns previously verified work into reusable computational structure without losing EBU accuracy or physical accounting.